אוניברסיטת תל-אביב

הפקולטה למדעי החיים ע"ש ג'ורג' ס. וייז

המדרשה לתארים מתקדמים

הנושא:

חקר היווצרותם של רצפים חוץ-כרומוזומליים בנמטודות מסוג *Caenorhabditis elegans*

חיבור זה הוגש כעבודת גמר לקראת התואר "מוסמך אוניברסיטה"
במסלול נוירוביולוגיה באוניברסיטת תל-אביב

על-ידי

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העבודה הוכנה במחלקה לנוירוביולוגיה של אוניברסיטת תל-אביב

בהנחיית

פרופסור עודד רכבי

אוגוסט 2023

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חתימת המנחה:

תאריך:

תוֹדָה

לחברים שלי שבזכותם סיימתי

לנעם ומאור שבשבילם סיימתי

לשרית, שצריך כמותה בכל ארגון

לעודד על ההזדמנות והתעוזה המחשבתית

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תקציר

מערכים חוץ-כרומוזומליים (ECA) נמצאים בשימוש נרחב במחקר של נמטודות מסוג *C. elegans*, אך עדיין קשה מאוד להעריך את אופן יצירתם ויציבותם. מחקרים חדשים מעלים את האפשרות שלרצפי דנ"א כאלה חשיבות בתהליכים ביולוגים שונים ובמיוחד בהתפתחות סרטן. במסגרת עבודת המוסמך, השתמשתי בטכנולוגיית ננו-פור (Nanopore sequencing) על מנת לחקור לעומק את המבנים הגנומיים של ECAs. יצרנו שני פלסמידים שונים המכילים רצפים שונים והשפעה פנוטיפית אחרת על מנת לחקור מה משפיע על הקשרים בין שברי הפלסמיד שבונים את ה-ECA. אימתנו את נוכחות ה-ECA באמצעות PCR, בודדנו DNA מתולעים המכילות את ה-ECA ורצפנו את ה-DNA באמצעות Nanopore sequencingניתחנו את הנתונים באמצעות כלים ביואינפורמטיים שפיתחנו במעבדה במיוחד עבור משימה זו וגילינו מספר מאפיינים של יצירת ECAs בנמטודה. המחקר שלנו מדגים שריצוף באמצעות טכנולוגיה זו יכול לספק כלי רב עוצמה ורב-תכליתי לחקירת יצירת ECAs ב-*C. elegans*. מצאנו ששני ECAs יכולים להתקיים במקביל באותה נמטודה אך נשארים נפרדים ואינם מתחברים ל-ECA אחד, ומאפיינים את יצירת המבנה של ECAs. העמקת ההבנה שלנו לגבי יצירת ECA בתולעת ומה משפיע עליה, יכולה להוות בעתיד בסיס לכלי משמעותי וחזק בביולוגיה סינטטית ליצירה של רצפי DNA ארוכים.

Abstract

Extrachromosomal arrays (ECAs) are widely used in *C. elegans* research (Reinke *et al.*, 2000), but their creation and stability can be challenging to assess. New studies raise the possibility that such DNA sequences are important in various biological processes and especially in the development of cancer. As part of my master's work, I applied nanopore sequencing to investigate the genomic structures of ECAs. We created two different plasmids containing two different phenotypic markers and characters in order to investigate what influences the connections between the plasmid fragments that built the ECA. We validated the ECAs presence by PCR, isolated DNA from worms containing the ECAs and sequenced the DNA using Oxford Nanopore Technologies (ONT) long-read sequencing. We analyzed the data using bioinformatics tools we developed in the lab specifically for this task. and discovered several characteristics of ECAs creation in the nematode. Our study demonstrates that nanopore sequencing can provide a powerful and versatile tool for investigating the creation of ECAs in *C. elegans*. We show that two ECAs can coexist in the same nematode but stay separate and do not combine to one ECA and characterize the structure creation of ECAs. Furthering our understanding of ECA creation in the worm can be a very powerful tool in synthetic biology for long DNA sequencing creation.

1. Introduction

1.1 *C. elegans* as a model organism

Caenorhabditis elegans (*C. elegans*) is a small, free-living roundworm that has become an important model organism in scientific research. Its popularity as a model organism can be

attributed to several key features, including its ease of cultivation, short generation time, transparent body, and well-defined nervous system. *C. elegans* has been extensively studied in fields such as developmental biology, genetics, neuroscience, and aging. The organism's entire genome has been sequenced (it was the first multicellular organism to have its genome sequenced), and it has been found to share a high degree of genetic similarity with humans, making it a valuable tool for studying various human diseases.

1.2 The extrachromosomal array phenomenon

For over three decades, it has been well-established that extrachromosomal array (ECA) can be generated by microinjection of DNA constructs into the gonads of adult worms (Stinchcomb *et al.*, 1985). ECAs can be up to 2-3M bp long and can pass down from parent to offspring. Not all progenies will inherit it, and the amount of ECA present in each cell is variable (Lin *et al.*, 2021). Previous studies utilized traditional techniques such as gel electrophoresis and karyotyping to examine this phenomenon. In this study, we employed nanopore sequencing, one of the most advanced sequencing technologies, to gain a better understanding of the mechanism behind ECA formation.

1.3 DNA repair mechanisms

The health of an organism and the reproductive success of a species rely on the integrity of the genome. The accurate transmission of genetic information is delicately balanced between genetic stability and mutational changes that serve as a source for evolutionary progress. To preserve the fidelity of the genome, it is imperative to synchronize surveillance and repair mechanisms (Sancar *et al.*, 2004).

Double-strand breaks (DSBs), pose a severe threat to genome integrity and can arise spontaneously or due to external factors like radiation exposure or chemotherapy. Cells have evolved two primary pathways to repair these breaks: NHEJ, or non-homologous end-joining, which directly connects the broken DNA ends, and HR, or homologous recombination, which uses homologous DNA sequences as templates for repair. The choice of repair pathway for DSBs affects the accuracy of the repair process, with HR being typically more precise, while NHEJ tends to result in errors, such as deletions or insertions at the repair site.

NHEJ is a swift and relatively uncomplicated mechanism that joins the two fractured ends of DNA directly, without necessitating homologous sequences. This process plays a crucial role in mending both pathological and physiological DSBs. If NHEJ malfunctions, it leads to heightened vulnerability to radiation and immunodeficiency. In comparison, HR is a more intricate but precise method that uses a similar DNA molecule as a model to reinstate the original sequence of the damaged DNA. HR is primarily active during late S phase or G2 phase when sister chromatids are accessible as models. HR plays a vital role in preserving genome stability and diversity through meiotic recombination and gene conversion. If HR fails, it results in chromosomal abnormalities and a predisposition to cancer (Lieber, 2010).

1.4 DNA sequencing technologies

DNA sequencing is the process of determining the order of nucleotides in a DNA molecule. It has many applications in biology, medicine, forensics and biotechnology. There are different types of DNA sequencing technologies that vary in their speed, accuracy, cost and throughput (Shendure *et al.*, 2017).

Some of the most common DNA sequencing technologies are:

- Sanger sequencing: the first method of DNA sequencing developed by Frederick Sanger and his colleagues in 1977. Sanger sequencing is accurate, reliable and widely used for small-scale projects such as gene cloning, mutation detection and microbial identification. However, it is slow, expensive and labor-intensive compared to newer methods (Goodwin, McPherson and McCombie, 2016).
- Next-generation sequencing (NGS): a collective term for various methods that use massively parallel sequencing to generate millions or billions of short reads from DNA fragments in a single run. NGS methods include Illumina sequencing, Roche 454 sequencing, and Ion Torrent sequencing (Goodwin, McPherson and McCombie, 2016). NGS methods have revolutionized genomic research by enabling high-throughput, low-cost and rapid analysis of whole genomes, transcriptomes, epigenomes and metagenomes. However, they also have some limitations such as short read lengths, high error rates for some platforms, high costs and complex data analysis (Hu *et al.*, 2021).
- Third-generation sequencing (TGS): an emerging term for methods that sequence single molecules of DNA without amplification or fragmentation. TGS methods include Pacific Biosciences (PacBio) SMRT sequencing and Oxford Nanopore MinION sequencing (Goodwin, McPherson and McCombie, 2016). TGS methods have the potential to overcome some of the challenges of NGS methods such as resolving complex genomic regions, detecting structural variations and identifying base modifications. However, they also face some challenges such as high cost per base, low throughput and high error rates.

1.5 Oxford Nanopore MinION Sequencing

Nanopore sequencing is a technique that works by passing nucleic acids through tiny protein pores called nanopores and then determining their nucleotide sequence from the variations in the electrical signal that they cause. Nanopore sequencing has several advantages such as real-time analysis, long reads, scalability, portability and direct sequencing of DNA or RNA molecules. However, it also faces some challenges such as high error rates and high cost per base (Patel *et al.*, 2018; Żmieńko and Satyr, 2020)

We chose to use this method because it can generate long reads of up to 100 kb or more, which can help resolve complex genomic regions- like our ECA. As a comparison, Sanger sequencing generates short reads of up to 1 kb, Illumina sequencing and MiSeq sequencing generate short reads of up to 300 bp.

As for accuracy, MinION has a high raw error rate of about 25-38%, which can affect the identification of single nucleotide variants and small indels. Sanger, Illumina and MiSeq sequencing have a low error rate of about 0.5-1%, which can provide more reliable base calling and variant detection (Neal-McKinney *et al.*, 2021). Our analysis algorithm took into account the high error rate of this method, as base calling errors on single nucleotides have a small impact on understanding the entire ECA structure.

2. Materials and methods

2.1 Nematode propagation

C. elegans worms were grown over Nematode Growth Medium (NGM) in 60mm Petri dishes. The worms were fed OP50 strain Escherichia coli. Well-fed nematodes cultured at 20°C were used in all experiments.

2.2 Strains

The following *C. elegans* strains were used: wild-type N2 Bristol, HT1593 *unc-119*(ed3) III , and the strains we created: BFF71 *unc-119*(ed3) III; pigEx9[pCFJ350 *egl-6p*::*ced-3*::*unc-119*], BFF155 pigEx27[rol-6p::rol-6::rol-6 3'UTR], BFF182 unc-119(ed3) III; pigEx9[pCFJ350 egl-6p::ced-3::unc-119] ;pigEx27[rol-6p::rol-6::rol-6 3'UTR]

2.3 Plasmids construction

2.3.1 pCF350

We started the experiment by using pCFJ350 - MCS(ttTi5605, II), from Erik Jorgensen lab (Addgene plasmid # 34866 ; http://n2t.net/addgene:34866 ; RRID:Addgene_34866). The plasmid contains *unc-119* gene, and its length is 12,597 bp.

2.3.2 pMAX

Since we intend to use both plasmids together, we wanted to design a plasmid that has no similarity to pCFJ350. For our backbone, we used the Gateway™ pDONR™221 vector from "ThermoFisher Scientific" to avoid ampicillin resistance and genes unknown for *C. elegans*. Our goal was to insert the *rol-6* gene, which is phenotypically observed, into the new plasmid, so we amplified it from pRF4 plasmid using PCR (Kramer *et al.*, 1990; Mello *et al.*, 1991). Our multiple attempts to insert *rol-6* into pDONR221 failed until we discovered that the backbone contained a toxic E. coli fragment (ccdB). After removing this toxic part by PCR, we ligated the plasmid on itself and created a plasmid that we can propagate (unlike pDONR™221). Then we added the *rol-6* fragment, resulting in pMAX (named after Maximillian Knott, a post doc in the lab, who was involved in the development of the plasmid). We validated the cloning by PCR and sanger sequencing.

2.3.3 Preparation of the plasmids

pCFJ350 and pMAX were extracted from the bacteria using QIAgene miniprep kit. We used restriction enzymes digestion to linearize the plasmids. pCFJ350 was digested with NehI and pMAX with EcoRV. We then cleaned them from endotoxins and conducted ethanol precipitation. After testing the concentration in nanodrop, we injected 100 ng/µl from each plasmid to the worms.

2.3.4 The differences between the plasmids

The first plasmid- pCFJ350 is about 12k base-pairs length, contains Ampicillin resistance, and an *unc-119* rescue gene. Before we injected it, it was digested with NehI, that linearized the plasmid and left it with sticky ends. Mutant worms (*unc-119*) were injected with this plasmid.

The second plasmid- pMAX, is about 8k bp length, contains Kanamycin resistance and a *rol-6* gene. This plasmid was linearized by EcoRV restriction enzyme that left the fragment with blunt ends, and then injected into WT worms (N2)

The “origin of replication” is the only sequence that both plasmids have in common.

2.4 Microinjection

Adult worms were immobilized in a drop of oil on a microscopy cover slip added with a 1.5% agarose pad. The injection was visualized under a Nikon Eclipse Ti microscope connected to an Eppendorf Femtojet microinjector. We loaded the needle with 2 µl of the injection mixture (water and 100 ng\µl of plasmid). The worms were injected in the distal arm (syncytium) of one of the two gonads. After injection, the worms recovered in M9 solution and placed on NGM

plates containing bacteria. The plates were stored at 20°C for 2-3 days, then phenotypes were observed under binocular vision to determine whether the progenies were positive for the ECA.

2.5 DNA extraction

N2 worms grown on a NGM plate were washed from the plate with M9 buffer and transferred to a 15 ml Falcon tube. The worms were then washed 3 times with M9 and 3 times with ddH20 and the supernatant was removed. The final wash involved half an hour in a hula-mixer. The supernatant was removed, and the tube was placed in the -80 incubator for at least 30 minutes. The frozen samples were then thawed at room temperature and grinded within the pestle tube. Genomic DNA extraction was then performed using the Monarch® HMW DNA Extraction Kit for Tissue from New England BioLabs. Worms were resuspended in lysis buffer and Proteinase K, and the tube was incubated overnight at 37°c within a hula-mixer. The following steps were performed according to the protocol provided by the manufacturer. The quality of the genomic DNA was assessed on a Nanodrop machine and Qubit, and the final concentration was adjusted to ~50ng/µL using ddH20. The length of genomic DNA was evaluated by gel electrophoresis using a 50kb ladder and 1% agarose. The genomic DNA was then stored at 4°C.

2.6 Nanopore sequencing

2.6.1 Library preparation

Preparation of sequencing libraries was performed according to the instructions provided by Oxford Nanopore Technology.

2.6.2 Ligation kit

We used SQK-LSK109, a ligation-based sequencing kit for processing the samples (Figure 1A). We chose this kit because it generates long reads that contribute to the understanding of the creation of very repetitive sequence we invest. During the protocol, ligase adds sequencing adapters on the fragments ends.

2.6.3 Rapid kit

As a control to ligase activity in the Ligation kit, we also used Rapid Sequencing Kit (SQK-RAD004). Instead of ligase, this kit uses transposase which simultaneously cleaves template molecules and attaches tags to the cleaved ends. Rapid Sequencing Adapters are then added

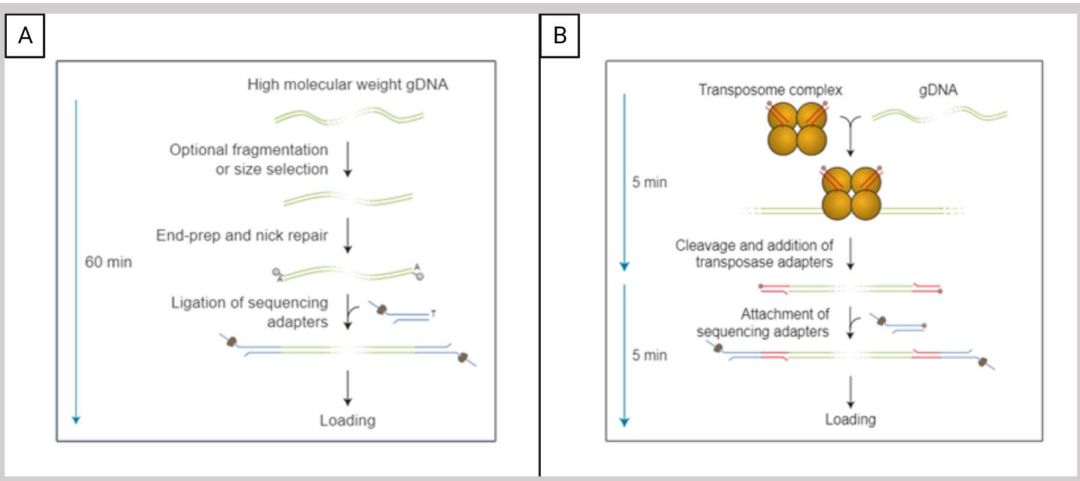


Figure 1: Nanopore sequencing kits

Schematic overview of library preparation for **A.** ligation kit **B.** rapid kit (Adapted from ONT website).

to the tagged ends (Figure 1B). We didn't use this kit in advance because the outcome reads are much shorter.

2.6.4 Preparation of MinION device and flowcell

All samples were run on a MinION device. Samples were loaded onto the FLO-MIN 106D flowcell (pore type: R9.4.1) purchased from ONT. The device was connected to a computer via USB and controlled by MinKNOW software. All sequencing experiments were generated in Yuval Ebenstein's lab.

2.7 Nanopore sequencing analysis

2.7.1 Quality control

The Nanopore FASTQ output files were filtered to remove low-quality reads (reads below a quality score of 8) using the Nanopore MinKNOW app. The files were then converted into the FASTA format.

2.7.2 Alignment

The sequence of each plasmid was aligned to the Nanopore reads using the following BLAST command (in orange- files names):

blastn -task 'dc-megablast' -max_target_seqs 2000000 -gapopen 2 -qcov_hsp_perc 5 -evalue 1e-5 -outfmt "6 qseqid sseqid slen pident length mismatch gapopen qstart qend sstart send evalue bitscore" -subject NANOPORE_READS_FASTA_FILE -query BUILDING_BLOCK_PLASMIDS -out OUTPUT_FILE.tsv

This command aligns the plasmid to all of the reads using the 'discountiguous megablast' (dc-megablast) algorithm, with a score penalty of 2 for gap openings, filtering out alignments that contain less than 5% of the plasmid's sequence, and filtering out alignments with e-value larger than 10^{-5}.

All downstream analysis was performed using Python code (Python version 3.8).

2.7.3 Read categorization

Next, the reads for which plasmid alignments were found were sorted into four categories: 'reliable' (reads that likely originate from an extrachromosomal array), 'plausible' (reads that are more likely than not to originate from an extrachromosomal array, but with lower confidence), 'spurious' (reads that may originate from an extrachromosomal array, but with very low confidence), and 'non-array' (reads that we have no evidence that originate from an extrachromosomal array).

Non-array reads were discarded from the analysis. All downstream analyses were performed on each of the three remaining categories separately.
Reads were sorted into the above categories based on the following criteria: read length; whether the read contains non-nematode sequences (plasmid components that were not flagged as endogenous); and % coverage of the read by plasmid fragments.

2.7.4 Junction analysis

Next, we extracted information about the alignment junctions within the reads in each category. A junction was defined as the sequence between two adjacent discovered plasmid alignments within the same read.

For each junction, we characterized it's length (the distance in bp between the end of the first alignment and the start of the second alignment), start/end coordinates of the first and second alignments in the junction, and the orientation of both alignments in the junction (whether they were aligned to the + or – strand of the plasmid).

Next, a control shuffled junction table was generated by shuffling the read IDs around, so that every read gets the same number of alignments it originally had, but the alignments are drawn randomly from all discovered alignments. The junction table was then generated using the method described above.

3. Experimental design and research goals

We started by injecting the two plasmids separately: pCFJ350 into *unc-119* strain (figure 2A, 1) and pMAX into *N2* strain (figure 2A, 2). We propagated the new strains- BFF71 and BFF155 (accordingly) until we had about 40 plates full of worms. Then, we extracted DNA (see methods) and sequenced it using Nanopore (figure 2B).

In this stage of the experiment, we wanted to understand the mechanism behind the creation of the ECA. Do the fragments tend to fuse in a certain orientation? Can we identify specific DNA repair mechanisms that are involved in the creation? Does the ECA contain endogenous parts that we didn't inject?

Additionally, we crossed the two strains we created; BFF71 and BFF155, resulting in BFF182 (figure 2A, 3). A progeny that phenotypically contains both arrays was identified and verified by PCR and it's DNA was extracted and sequenced as before.

As a result of this cross, we were able to answer the following questions regarding ECAs; did the two ECAs fuse together? did they stay separately? Is it even possible for a worm to have both at the same time?

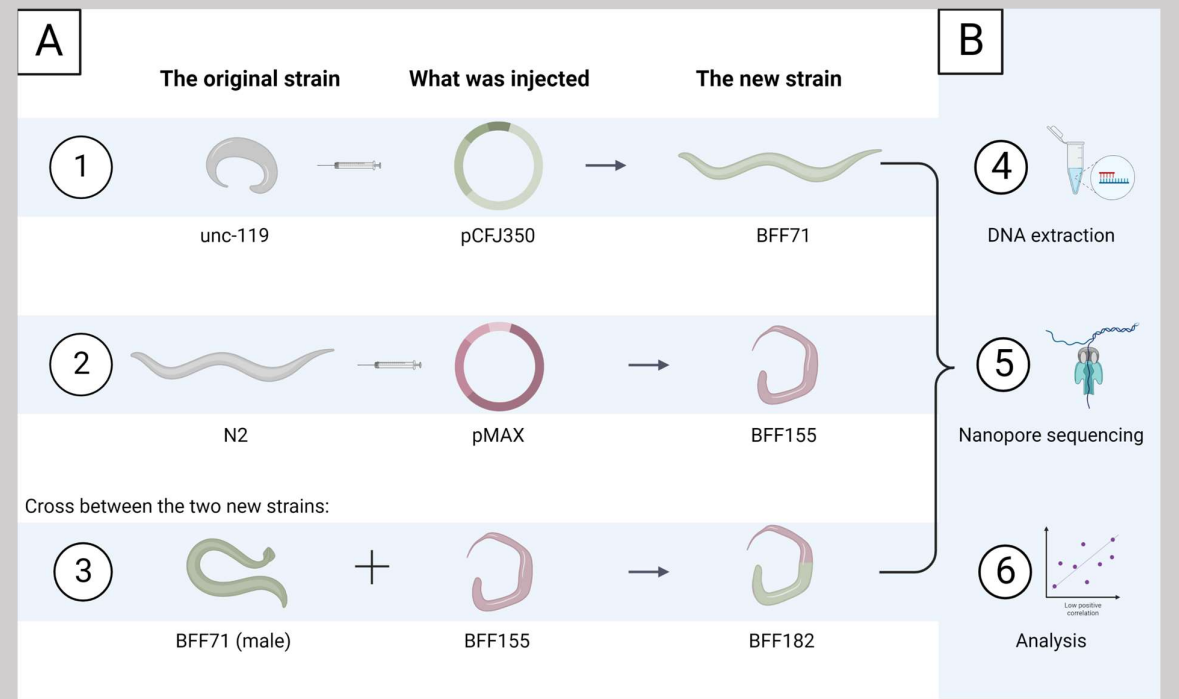


Figure 2: experiment design

Schematic overview of the experiments we conducted. **A.** Injecting pCFJ350 into *unc-119* worms (1), and pMAX into N2 worms (2). After we had the two new strains, we crossed between them (3) and created BFF182 **B.** As soon as we propagated the three strains, we extracted their DNA, sequenced it using Nanopore, and analysed the results.

4. Results

4.1 BFF71 ligation kit

We sequenced the DNA from BFF71 worms. The total throughput of the sequencing run was 8.49 Gbp, with 7.53 Gbp passing quality control (see methods). The estimated N50 of the sequencing run was 16.06 Kbp.

4.1.1 Reads analysis

Before analyzing the data, we wanted to gain a better understanding on what happened during the ECA creation. The plasmid map (Figure 3A) was converted to a colour map (Figure 3B). We selected a cyclic colour map so we can easily distinguish between continuous parts in the ECA, which are comprised of whole fragments (12 Kbp) connected by sticky ends or other connections that are not whole or continuous. Our pipeline (see methods) generated visual

reads (Figure 3C) that can infer the structure of the ECA. For instance, in Read 5, we observed two whole fragments connected to each other, followed by a few broken fragments.

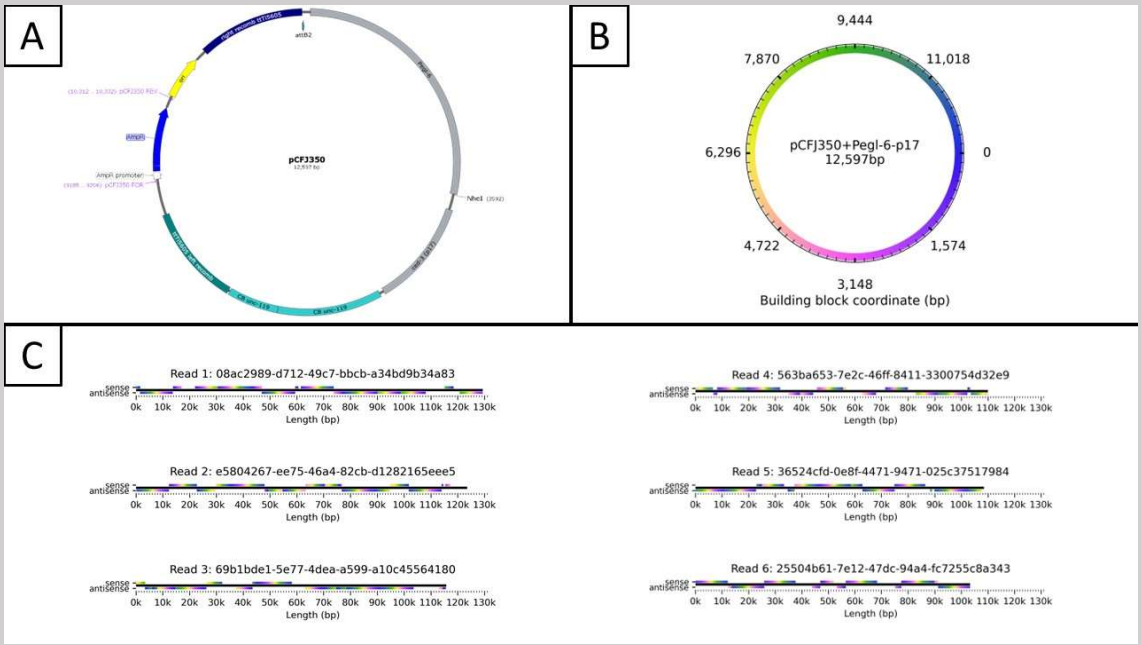


Figure 3: Analysis of BFF71 Sequencing Results with Ligation Kit

A. pCFJ350 plasmid map (from SnapGene). The plasmid contains ampicillin resistance and *unc-119* gene. Also in the map: NehI restriction site, and pCFJ350.FOR/REV the primers we used to validate this plasmid presence. **B.** pCFJ350 color map. **C.** The longest six reads from this sequencing.

4.1.2 Junction analysis

Next, we wanted to analyze the connections between fragments and see if there was a preference for a particular connection type, indicating a particular DNA repair mechanism. The algorithm tested each junction (a connection between two fragments) labelling it according to the orientation of the fragment before and after the junction and dividing the data into four groups of connections: sense-antisense (s-a), antisense-sense (a-s), sense-sense (s-s), and antisense-antisense (a-a). The determination of sense or antisense is arbitrary but consistent. We have seen that there is a small bias towards junctions of the same orientation (s-s, a-a) (Figure 4A). We wanted to check whether the fragments' length is related to this bias, therefore we chose to analyse only fragments above 11Kbp (Figure 4B) and we saw that the bias still exists.

To better understand the difference between the four groups (a-a, s-s , a-s, s-a), we also looked at the coordinates of the junctions. Each junction was plotted on a scatter plot (Figure 4C) using three parameters: The final coordination of the fragment before the junction (predecessor, X-axis), the starting coordination of the fragment after the junction (successor, Y-axis), and the mean lengths of the two fragments of the junction (the color of the dot). All the data were divided into the four groups of junction type (four different charts) so we could examine if there were different trends between them. As a control, we shuffled all the points (supplementary B) and found that there were a lot of "hidden" points that probably hide one

after another. We decided to make the same scatter plot again, while the colour will reflect the density of points on the plot instead of fragments length (Figure 4D).

Upon analyzing the two scatter plots together, we can observe several trends as represented by the few circles and lines on the graphs. Firstly, most of the fragments tend to connect to other fragments by keeping the original order from the plasmid (the end coordinate of one fragment connect to the start coordination of the next fragment), as indicated by the purple dots in Figure 4D, suggesting the influence of the sticky ends. Secondly, there are instances where whole fragments connect to broken fragments, represented by the red lines in Figure 4C. Thirdly, broken fragments are more likely to connect with the same coordination, as shown by the yellow line in Figure 4C. Finally, there are some random connections as indicated by the green circle in Figure 4C.

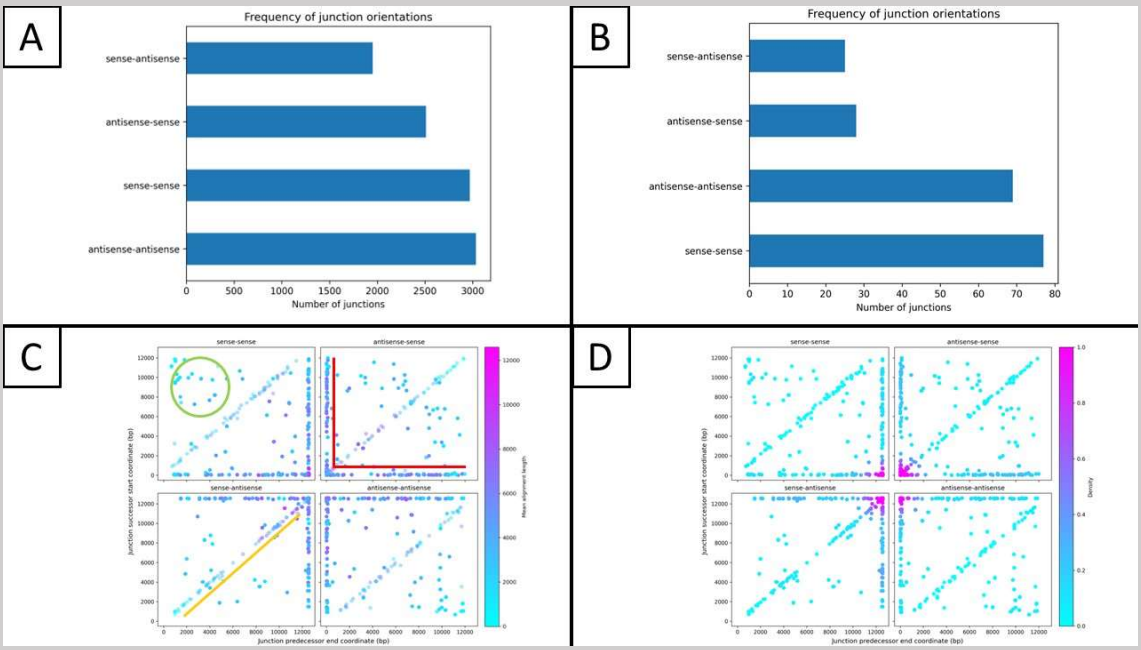


Figure 4: Junctions Analysis of BFF71 Sequencing Results with Ligation Kit

A. The frequency of junction orientation, divided into four groups (a-a, s-s, s-a, a-s). **B.** Frequency of junction orientation only for fragments of length 11Kbp and above. **C.** Scatter plot of fragment junctions, where the X-axis represents the final coordinate of the predecessor fragment and the Y-axis represents the starting coordinate of the successor fragment. The color of the dots represents the mean length of the two fragments, with bright blue indicating the shortest and bright purple indicating the longest. Red, yellow, and green circles/lines mark specific trends as described in the text, including whole fragments connected to broken fragments, broken fragments connecting with the same coordination, and random connections. **D.** Same as C, but the color of the dots represents the density of points on the plot, with bright blue indicating low density and bright purple indicating high density.

4.2 BFF155 ligation kit

We sequenced the DNA from BFF155 worms. The total throughput of the sequencing run was 12.62 Gbp, with 9.59 Gbp passing quality control (see methods). The estimated N50 of the sequencing run was 8.89 Kbp.

4.2.1 Reads analysis

At this sequencing, the plasmid map (Figure 5A) was also converted to a colour map (Figure 5B), but we didn't choose a cyclic map, because there are no sticky ends that were suspected to influence the connectivity. By looking briefly at the reads (Figure 5C) we can see that there are less connections from different orientation- a thing that we wanted to check in the data.

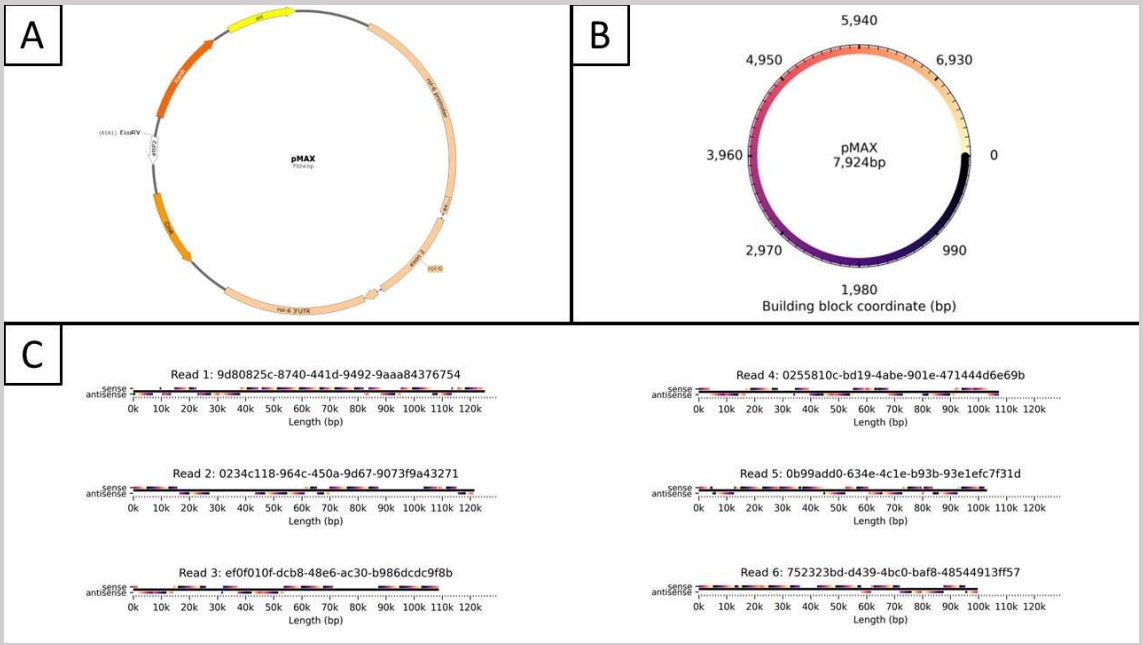


Figure 5: Analysis of BFF155 Sequencing Results with Ligation Kit

A. pMAX plasmid map (SnapGene). The plasmid contains kanamycin resistance and *rol-6* gene. Also in the map: EcoRV restriction site. **B.** pMAX color map. **C.** The longest six reads from this sequencing.

4.2.2 Junction analysis

As suspected, there was a bias towards connections at the same orientation (s-s, a-a) in BFF155 strain (Figure 6A) compared to the previous strain (Figure 4A). This bias was also present between whole fragments (Figure 6B) when we filtered the data to only include fragments of 6 Kbp and above.

By examining the two scatter plots together (Figure 6C and 6D), we noticed some interesting trends:

In plot C, we observe several trends among the junctions. Broken fragments tend to connect to the same coordination, as shown by the blue line, indicating that they tend to re-join with their original directionality. There are also many random connections shown by the green circle. Furthermore, for junctions between fragments of the same orientation (s-s or a-a), most fragments connect to each other at the digesting site (coordinate 0 and 8000) despite the absence of sticky ends at the edges of the fragments.

In plot D, we see the same junctions as in plot C, but the colour of the dots now indicates the density of the junctions. We can see that there are many junctions between fragments of the same orientation and length, bright yellow clusters. There are also unexpected accumulations of junctions between fragments of opposite orientation (s-a or a-s) in the 1000-2000 and 7000

coordinate range, as shown by the purple circles. It is important to note that these unexpected junctions are in the s-a or a-s orientations that are less common than the s-s and a-a junctions in this strain (Figure 6A).

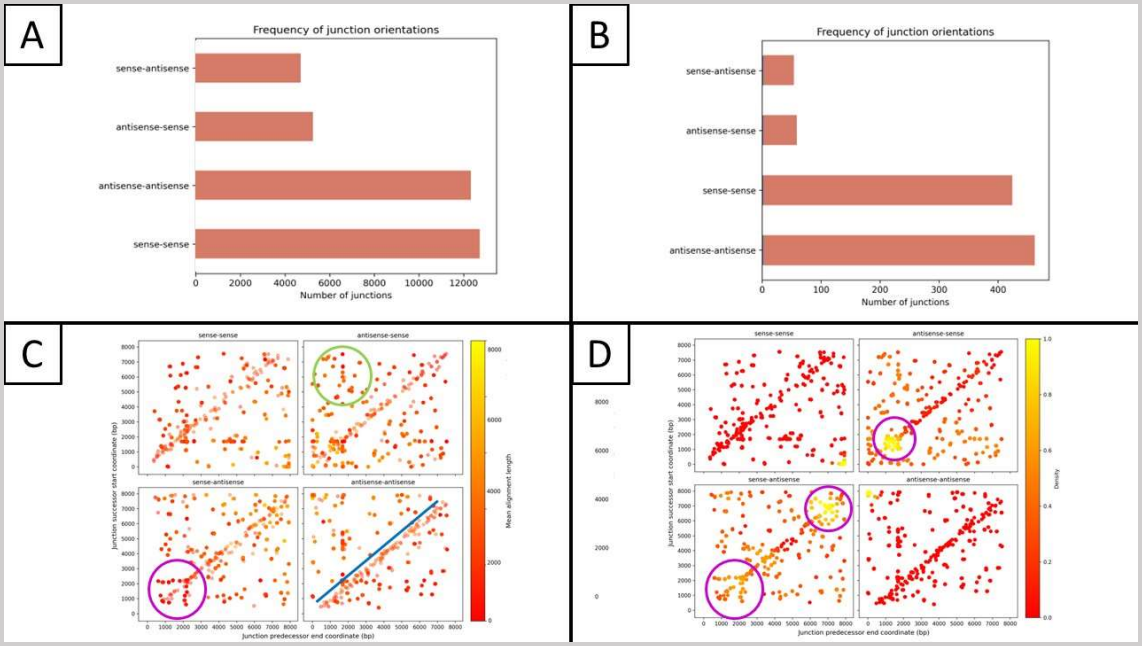


Figure 6: Junctions Analysis of BFF155 Sequencing Results with Ligation Kit

A. Frequency of junction orientation, divided into four groups (a-a, s-s, s-a, a-s). **B.** Frequency of junction orientation only for fragments of length 11Kbp and above. **C.** Scatter plot of fragment junctions, where the X-axis represents the final coordinate of the predecessor fragment, and the Y-axis represents the starting coordinate of the successor fragment. The color of the dots represents the mean length of the two fragments, with red indicating the shortest and yellow indicating the longest. Blue, green, and purple circles/lines mark specific trends as described in the text. **D.** Same as C, but the color of the dots represents the density of points on the plot, with red indicating low density and yellow indicating high density.

4.3 BFF182 creation

Our objective was to investigate the coexistence and potential crossover between two ECAs within a single worm. To achieve this, we conducted a cross between two strains, BFF71 males and BFF155 hermaphrodites (Figure 7A), resulting in a strain with three distinct traits: homozygote to *unc-119* mutation, containing pCFJ350 plasmid (rescue to the mutation), and containing pMAX plasmid (roller phenotype). Four days after the cross, F1 worms with a roller phenotype were separated into numbered plates. At this stage we had heterozygote to *unc-119* mutation rollers (Figure 7A, showed in blue) or no *unc-119* mutation rollers (Figure 7A, showed in red). After allowing all the rollers to lay eggs, we verified the presence of pCFJ350 plasmid with PCR. F2 rollers were then separated from pCFJ350 positive plates (Figure 7B) and given time to lay eggs, during which we confirmed the homozygous state of *unc-119* genomic mutation with PCR. Phenotypically distinguishable plates with uncoordinated worms were selected, with all the worms being homozygous to the *unc-119* mutation. Phenotypically Wild-Type worms contained only pCFJ350, uncoordinated worms contained only pMAX or no plasmid at all, and roller worms contained both plasmids and thus formed the new strain, BFF182, with a distribution of 45.1% WT, 27.45% uncoordinated, and 27.45% rollers (Figure 7C). F3 worms were sequenced and verified for both plasmids through PCR to confirm the presence of *unc-119* mutation (Figure 7D).

4.4 BFF182 ligation kit

Subsequently, we performed DNA sequencing on BFF182 worms. The sequencing run generated a total throughput of 2.38 Gbp, with 2.09 Gbp passing quality control (see methods). The estimated N50 of the sequencing run was 15.34 Kbp. Despite the limited amount of data, we were able to achieve our primary objective of understanding the behavior of the two ECAs during a cross. However, due to the small data size, we were unable to compare the results to our previous sequencing and analyze the junctions.

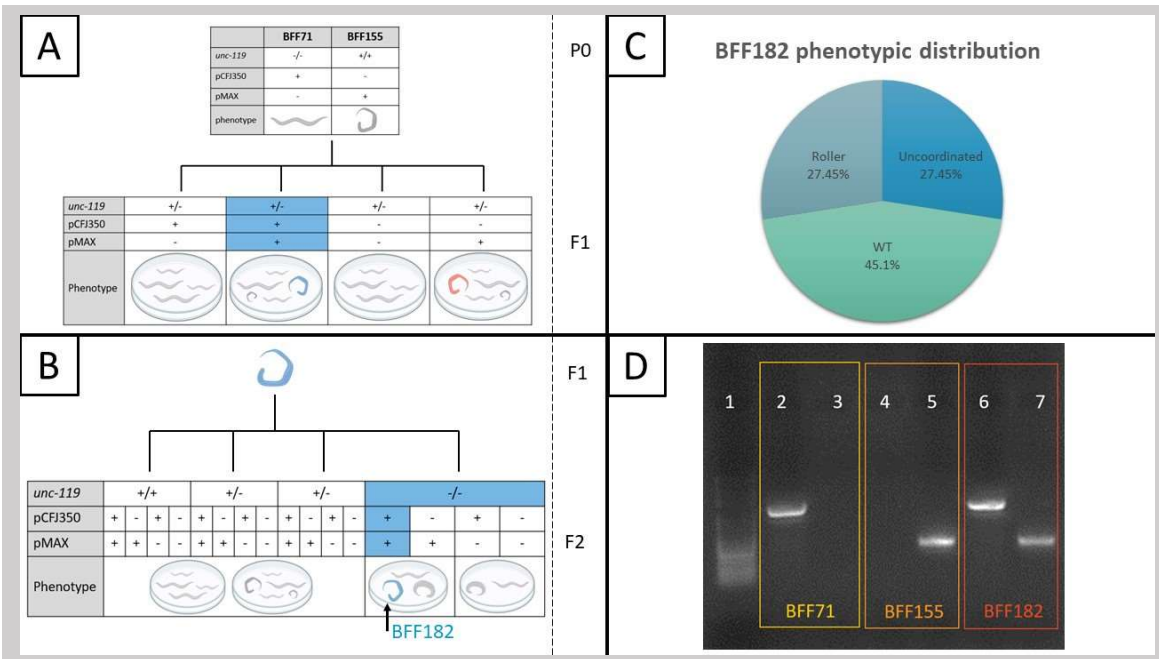


Figure 7: BFF182 strain creation

A. Schematic representation of the cross between BFF71 and BFF155 strains, detailing the genotype and phenotype of the starting strains (P0) and potential progenies in the F1 generation. Blue indicates worms chosen after PCR validation, while red indicates additional rollers to be separated in the next phase. **B.** The potential progenies in the F2 generation starting with the selected F1 worms. Blue indicates the new strain, BFF182, which contains both plasmids and the genomic mutation. **C.** The phenotypic distribution of the BFF182 strain. **D.** Gel electrophoresis image showing PCR validation for the presence of the two plasmids. Lane 1: 100Kb ladder. Lanes 2 and 3: BFF71 strain. Lanes 4 and 5: BFF155 strain. Lanes 6 and 7: BFF182 strain. Primers for pCFJ350 were used in lanes 2, 4, and 6, while primers for pMAX were used in lanes 3, 5, and 7.

4.4.1 Reads analysis

We applied the same color maps used in our previous analyses: BFF71 with a cyclic color map (Figure 3B), and BFF155 with a linear color map (Figure 5B). Our goal was to identify reads containing fragments from both ECAs, which would serve as evidence for their fusion. We created mock reads for every optional read we could have got in this sequencing (Figure 8A) and then, analyzed all 2,698 reads and found no instances where fragments from both ECAs were present in the same read. We have provided examples of the first 9 reads in Figure 8B. While our sequencing data is limited, these results suggest that the two ECAs did not undergo significant crossover during the cross.

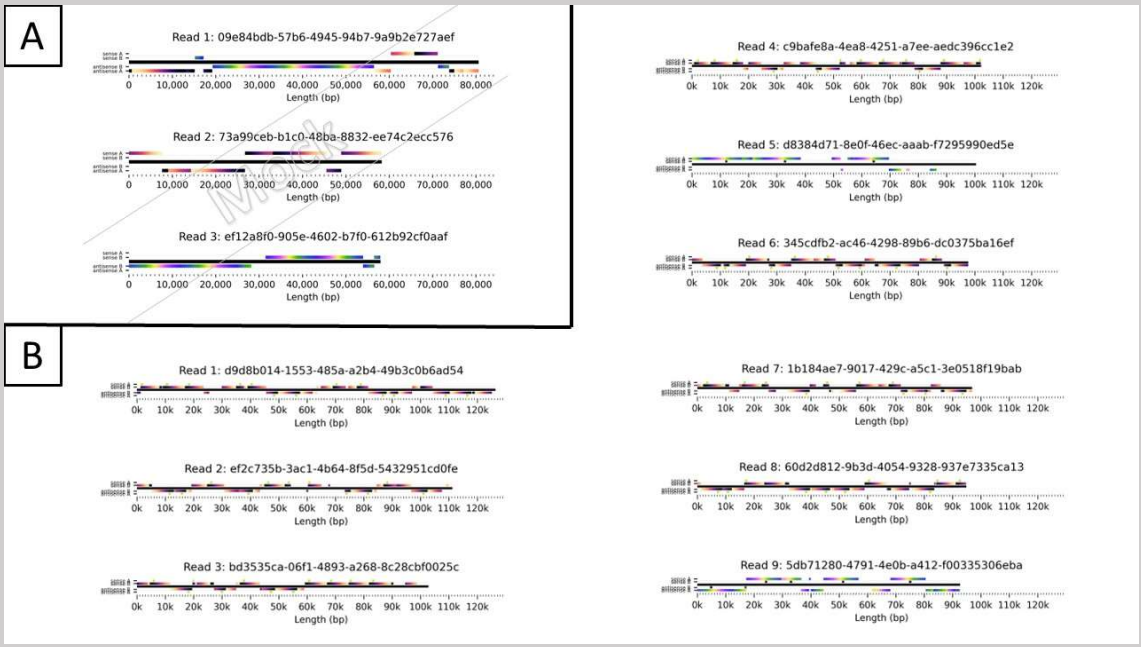


Figure 8: Analysis of BFF182 Sequencing Results with Ligation Kit

A. Schematic representation of a potential read in BFF182, demonstrating a crossover event between the two extrachromosomal arrays (ECAs) from the parent strains BFF71 and BFF155. **B.** The first 9 reads from a total of 2,698 reads were analyzed from BFF182 sequencing results. No reads were found to contain fragments from both ECAs, indicating a lack of evidence for a crossover event.

Note: The results were generated using the same color maps as in previous figures, with BFF71 represented by a cyclic color map (Figure 3B) and BFF155 represented by a linear color map (Figure 5B).

4.5 BFF71 rapid kit

We wanted to make sure that the ligase that we used in the ligation kit, didn't affect the connections between the fragments and on the gaps between them. As a control we used a different library preperation kit- Rapid kit (see methods) that doesn't contain ligase in its protocol. We sequenced the DNA from BFF71 worms. The total throughput of the sequencing run was 3.45 Gbp, with 2.58 Gbp passing quality control (see methods). The estimated N50 of the sequencing run was 11.91 Kbp.

As we can see (Figure 9), the graphs from the Rapid kit sequencing (Figure 9D, 9E, 9F) are very similar to the Ligase kit sequencing (Figure 9A, 9B, 9C), confirming that the use of the ligation kit did not affect the connections, junctions and gaps reflecting on the reads.

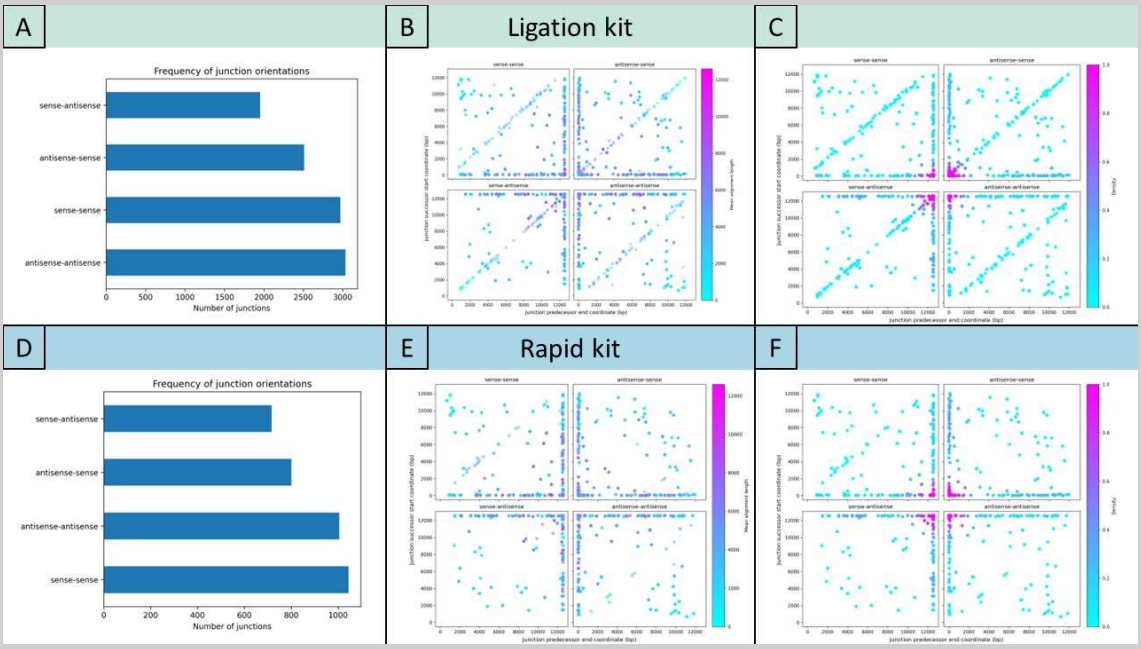


Figure 9: Analysis of BFF71 Sequencing Results with Rapid Kit Vs. Ligation Kit

A, B, C. Sequencing Results with Ligation Kit. **D, E, F**. Sequencing Results with Rapid Kit.

A, D. The frequency of junction orientation, divided into four groups (a-a, s-s, s-a, a-s). **B, E.** Scatter plot of fragment junctions, where the X-axis represents the final coordinate of the predecessor fragment, and the Y-axis represents the starting coordinate of the successor fragment. The color of the dots represents the mean length of the two fragments, with bright blue indicating the shortest and bright purple indicating the longest. **C, F.** Same as B and E, but the color of the dots represents the density of points on the plot, with bright blue indicating low density and bright purple indicating high density.

5. Discussion

In this study, we investigated the creation of extrachromosomal arrays (ECAs) in *C. elegans* nematodes. This phenomenon is well known and well used, but poorly understood. We wanted to use long read sequencing to look at the creation process of the ECA sequence and not only at the phenotypic effects.

In order to do so, we created two different plasmids (pCFJ350 and pMAX), injected them into two different strains (*unc-119* and N2 accordingly) and by that, created two new strains (BFF71 and BFF155 accordingly). We designed these plasmids to answer as many questions as we can, so in one injection we could understand the influence of multiple factors on ECA creation.

Looking at the results, the main difference we saw between the strains was on the junction's orientation frequency. On BFF71 (Figure 4A) we see almost a uniform distribution. It makes sense because pCFJ350 has sticky ends. Fragments with sticky ends are not supposed to prefer a specific orientation because one fragment's sticky end is compatible with the sticky end of a second fragment on both sides. However, there is a small bias towards junctions of the same orientation. We thought that this tendency was due to the broken fragments tending to bind to each other in the same orientation through the mechanism of Homologous Recombination (HR). Our hypothesis was that if we consider only whole fragments (12 Kbp) containing the sticky ends, the bias will disappear. When analyzing only fragments above 11Kbp (Figure 4B), we saw, surprisingly, that the

bias still exists. However, this bias is due to only ~200 fragments out of thousands, so it can be ignored.

On the other hand, on BFF155 (Figure 6A) there is a strong bias toward connections on the same orientation. This makes sense because blunt ends (like in pMAX), in order to connect using HR, have to be in the same orientation.

It will be likely to assume that the difference between the two strains is due to the end type of the fragments (sticky or blunt). Fragments with sticky ends (BFF71, Figure 4A) have less bias toward connections on the same orientation because they have an additional way of connecting (through the sticky ends) that the fragments with blunt ends do not.

Our results suggest that the type of the sequence's end (sticky or blunt) can influence ECA creation, although we were not able to directly compare the effect of each end type due to our experimental design. In retrospect, we should have injected the same plasmid twice, with different types of ends only (unfortunately, these are complicated experiments, and each run takes months to complete).

Another difference that we observed is that in BFF71 we saw instances of whole fragments connected to broken fragments (red line in Figure 4C). In BFF155 however, we did not observe these connections. This might suggest that the presence of sticky ends, even if only on one of the two fragments, helps connect the two fragments.

Besides the differences between the two plasmids, we have also seen similar trends: Broken fragments tend to connect in the same coordination. It can be seen by the presence of the diagonal lines in the figures 4C, 4D and 6C, 6D, suggesting that **HR mechanism is involved in the creation of ECAs.** Additionally, when looking at the density plots (Figures 4D and 6D), the most common connection is "end to start" (the end coordinate of the predecessor connects to the start coordinate of the successor), like the original structure of the uncut circular plasmid. Why would this be the case in BFF155? How does the plasmid "know" to connect "end to start" in the absence of sticky ends, and not "end to end" or "start to start"? We have one possible answer to this question; as the restriction reaction efficiency is not 100%, we might have injected circular plasmids, that could have served as templates for HR, resulting in mostly "end to start" connections.

Another understanding we wanted to gain is what happens to two different ECAs after a cross between two strains. We successfully isolated worms with the two ECAs, confirming that **two ECAs can coexist in the same individual nematode** (Figure 7D) . Further, **the two ECAs did not combine**, but rather stayed separately, suggesting that after its creation, the ECA is rather stable. This result might have been different had we injected the two plasmids simultaneously to create the strain as opposed to crossing two strains together. Moreover, the two ECAs carry different genes that are expressed in different tissues (muscle tissue for *rol-6* and neurons for *unc-119*). Maybe if the plasmids had expressed functionally similar genes or genes that are expressed in the same tissue, we would have got a different outcome.

Future directions:

Our study suggests that DNA repair mechanisms play a role in ECA creation, but further work is needed to understand the specific contributions of individual mechanisms or genes. It would be interesting to address this by injecting the same plasmids to worms carrying mutated alleles in DNA repair genes and comparing their ECA formation to wild-type nematodes.

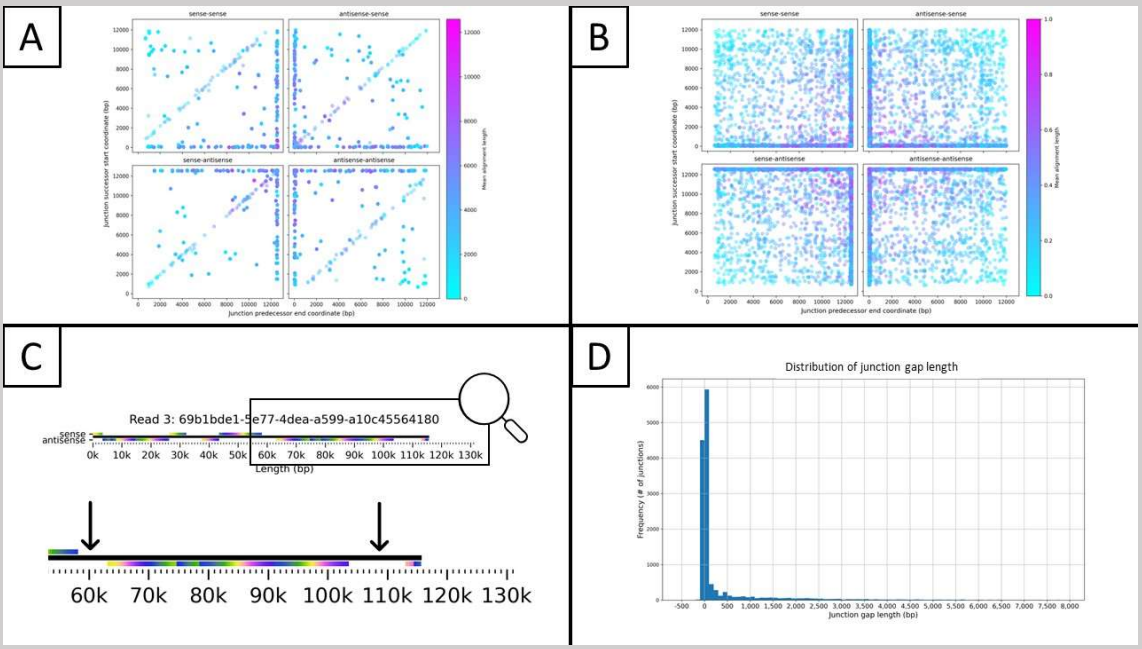
Our experimental design can also serve to study RNA interference (RNAi). Endogenous RNAi requires the transcription of double-stranded RNA (dsRNA). The presence of repetitive sequences in ECAs, some of them connected in a sense-antisense junction, can create a dsRNA structure when transcribed. Such structures can hypothetically serve as precursors that could later be cut by DCR-1 into primary siRNAs, and initiate silencing of the genes expressed by the ECA. Thus, it would be interesting to look for worms that carry the array (validated by PCR) but do not express it (using the phenotypic effects, e.g., uncoordinated), and see whether in these worms, there are higher occurrences of sense-antisense or antisense-sense junctions.

Intriguingly, we have found sequences in the ECAs that do not align to the injected plasmids (supplementary C and D). It would be interesting to study these sequences and look for their origin, maybe they can shed more light about the biological role of the ECA inside the worms. For example, if these parts are endogenous sequences, it may have evolutionary consequences because it means that the mechanism that creates ECAs can 'cut and paste' sequences from the genome, or maybe these sequences originate from pathogens, which might suggest that the ECA is part of the immune system of the worms.

Finally, we propose a future direction for investigating ECA creation that involves injecting fragments of DNA with different long overlaps between them (2000/ 1000/ 500/ 100 bp). By analysing the resulting arrays, we can determine the minimum number of base overlaps required to guarantee specific connections. With this knowledge and more, we can theoretically design a small synthetic chromosome with a length of 2-3 million bases, which would be much cheaper than the current industry standard, and easily maintained and amplified in *C. elegans* worms.

In summary, our study illuminates the intricate process of ECA formation in *C. elegans*. Moreover, it identifies potential future research directions aimed at unraveling the underlying molecular mechanisms and exploring the possible applications of this phenomenon within the field of synthetic biology.

6. Supplementary Figures



Supplementary BFF71 Sequencing Results with Ligation Kit

A. A scatter plot of fragment junctions, already shown on Figure 4A. **B.** Shuffled data of A. We found out lots of "hidden" dots that hinted to us on dots accumulation areas. **C.** Enlargement of part of one of the reads. Arrows points on "gaps" in the alignment. **D.** Analyzing the gaps we found on the alignment. The length of the gap mention on the X-axis, while the Y-axis represents the frequency of such gap length in the data.

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GEORGE S. WISE FACULTY OF LIFE SCIENCES

GRADUATE SCHOOL

Understanding the creation of extrachromosomal arrays in *Caenorhabditis elegans* nematodes.

Thesis submitted towards the M.Sc. degree in Neurobiology.
at Tel-Aviv University

by
Aviya Zevuloni Weiser

The research was performed in the Department of Neurobiology

under the supervision of

Professor Oded Rechavi

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